Arjuna JEE 2025

Mathematics

Permutations and Combinations

DPP: 7

Q1 Out of seven consonants and four vowels, the number of words of six letters, formed by taking four consonants and two vowels is (Assume that each ordered group of letters is a word):

(A) 210 (B) 462

(C) 151200 (D) 332640

Q2 Twelve persons are to be arranged around two round tables such that one table can accommodate seven persons and another five persons only. The number of ways in which these 12 persons can be arranged is

(A) ${}^{12}\mathrm{C}_5 7!4!$

(B) $6!4!$

(C) ${}^{12}\mathrm{C}_5 6!4!$

(D) none of these

Q3 Twelve persons are to be arranged around two round tables such that one table can accommodate seven persons and another five persons only. The number of ways of arrangements if two particular persons A and B do not want to be on the same table is

(A) ${}^{10}\mathrm{C}_4 6!4!$

(B) $2.\,{}^{10}\mathrm{C}66!4!$

(C) ${}^{11}\mathrm{C}_6 6!4!$

(D) none of these

Q4 Twelve persons are to be arranged around two round tables such that one table can accommodate seven persons and another five persons only. The number of ways of arrangement if two particular persons A and B want to be together and consecutive is

(A) ${}^{10}C_7 6!3!2! + {}^{10}C_5 4!5!2!$

(B) ${}^{10}C_5 6!3! + {}^{10}C_7 4!5!$

(C) ${}^{10}C_7 6!2! + {}^{10}C_5 5!2!$

(D) none of these

Q5 The number of ways of selecting 4 letters from the letters of the word 'SYLLABUS' such that two letters are distinct and two letters are alike, is

(A) 22 (B) 19

(C) 20 (D) 21

Q6 Total number of ways of forming 6 letter word from the letters of word "PROPORTION".

(A) 11130 (B) 1140

(C) 11120 (D) 11150

Q7 The number of four-letter words that can be formed using the letters of the word 'FAILURE', so that F is included in each word, is

(A) ${}^{6}\mathrm{C}_4 \times 4\,!$

(B) ${}^{6}\mathrm{C}_3 \times 4\,!$

(C) ${}^{6}\mathrm{C}_4$

(D) ${}^{6}\mathrm{C}_4 \times 3\,!$

Q8 The number of 5 lettered words formed with the letters of the word CALCULUS is divisible by :

(A) 2 (B) 3

(C) 5 (D) 7

Q9 The number of five-letter words containing 3 vowels and 2 consonants that can be formed using the letters of the word 'EQUATION' so that the two consonants occur together, is

(A) 720 (B) 1440

(C) 240 (D) 480

Q10 If the four-letter words (need not be meaningful) are to be formed using the letters from the word "MEDITERRANEAN" such that the first letter is R and the fourth letter is E, then the total number of such words, is

(A) $\frac{11!}{(2!)^3}$
(B) 59
(C) 110
(D) 56

Q11 Harsh invites 13 guests to a dinner and places 8 of them at one table and remaining 5 at the other, the table being round. The number of ways he can arrange the guests is

(A) $\frac{11!}{40}$
(B) $9!$
(C) $\frac{13!}{40}$
(D) $\frac{12!}{40}$

Q12 The number of words of 4 letters which can be formed with the letters of the word MATHEMATICS is α, then $\frac{\alpha}{300}$ is

(A) 7.17 (B) 8.18
(C) 9.19 (D) 10.1

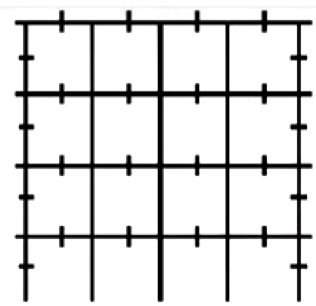
Q13 There are n points in a plane of which p points are collinear. How many lines can be formed from these points

(A) ${}^{(n-p)}C_2$
(B) ${}^{n}C_2 - {}^{p}C_2$
(C) ${}^{n}C_2 - {}^{p}C_2 + 1$
(D) ${}^{n}C_2 - {}^{p}C_2 - 1$

Q14 The straight lines l_1, l_2, l_3 are parallel and lie in the same plane. A total number of m points are taken on l_1, n points on l_2, k points on l_3. The maximum number of triangles formed with vertices at these points, are

(A) ${}^{m+n+k}\mathrm{C}_3$
(B) ${}^{m+n+k}\mathrm{C}_3 - {}^{m}\mathrm{C}_3 - {}^{n}\mathrm{C}_3 - {}^{k}\mathrm{C}_3$
(C) ${}^{m}\mathrm{C}_3 + {}^{n}\mathrm{C}_3 + {}^{k}\mathrm{C}_3$
(D) none of these

Q15 The number of squares in given figure, is



(A) 8 (B) 18
(C) 20 (D) 40

Q16 There are n concurrent lines and another line parallel to one of them. The number of different triangles that will be formed by the $(n+1)$ lines, is

(A) $\frac{(n-1)n}{2}$
(B) $\frac{(n-1)(n-2)}{2}$
(C) $\frac{(n+1)n}{2}$
(D) $\frac{(n+1)(n+2)}{2}$

Q17 5 Indian and 5 American couples meet at party and shake hand. If no wife shake hand with her own husband and no Indian wife shakes hands with a male, then the number of handshakes that take place in the party is $15.k$, then the value for k is

Answer Key

Q1	(C)
Q2	(C)
Q3	(B)
Q4	(A)
Q5	(C)
Q6	(A)
Q7	(B)
Q8	(A, B, C)
Q9	(B)
Q10	(B)
Q11	(C)
Q12	(B)
Q13	(C)
Q14	(B)
Q15	(C)
Q16	(B)
Q17	9